Thiago Vasco

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EDUCATION

University of British Columbia — Expected Graduation: 05/26

Master of Science, Mathematics, Institute of Applied Mathematics (IAM), Cumulative Average 88.4%

- Developed novel diagnostic tools for industrial control systems in partnership with the Parkland Corporation, and completed my thesis *Automatic Diagnosis of Industrial Model Predictive Controllers with Steady-State Optimizers*
- Co-supervised by Prof. Philip Loewen and Prof. Bhushan Gopaluni

University of British Columbia — Graduated: 12/23

Bachelor of Science, Major in Statistics, Minor in Mathematics, Cumulative Average 85.3%

- UBC Centennial Scholars Entrance Award, Dean's Honour List, British Columbia Government Scholarship, Professor Richard Lipsey Award in International Economics

EMPLOYMENT

Intern Analyst, Applied Research & Models, CPP Investments, Toronto ON — 05/22 – 08/22

- Developed knowledge of professional software engineering practices for a full software development cycle, including continuous integration, coding standards, and code reviews
- Added functionality to analyze factor and cash exposures within asset allocation software
- Helped research a novel factor investing method which better characterizes idiosyncratic risk of illiquid asset classes
- Handled other ad hoc requests such as data cleaning and updating software packages

Intern Analyst, Portfolio Design & Construction, CPP Investments, Toronto ON — 01/22 – 04/22

- Debugged and added functionality to experimental dynamic portfolio optimization model created by the team
- Created a high-quality GUI using R Shiny for researching optimal factor exposures under different model assumptions

Project Assistant, CANARY Cognition Research Lab (UBC), Vancouver BC — 05/21 – 08/21

- Developed software for training and testing of Hugging Face transformer models applied to NLP classification, and contrasted performance with simpler models and other types of neural architectures
- Wrote code to parse language data and accomplish other preprocessing tasks

CLASS PROJECTS (available on my GitHub page)

Missile Guidance via Recursive LQR (MECH 509 – Control Theory)

- Developed a control strategy for navigating a 2D missile towards moving aerial targets; the missile is modeled as a nonlinear time-varying system, and the control strategy recursively solves a differential Riccati equation every time the predicted trajectory of the aerial target is updated
- Implemented the control system in Python, and ran simulations using a publicly available SDE solver

Primal-Dual Interior-Point Optimization with Application to MPC (CPSC 536M – Convex Analysis and Optimization)

- Implemented a primal-dual interior-point algorithm for solving discrete-time optimal control problems in Julia
- The solver can handle nonlinear system dynamics, and makes use of various techniques such as Mehrotra's predictor-corrector, merit functions, and damped BFGS updates for estimating the Hessian matrix of the Lagrangian

Model-Based Reinforcement Learning Using Predictive Sampling MPC (CPSC 533V – Learning to Move)

- Implemented a model-based RL algorithm using PyTorch, where system dynamics is approximated using a neural network, and optimal actions are computed using a sampling-based model predictive controller

ADDITIONAL INFORMATION

- **Legal Status:** Canadian Citizen
- **Academic Interests:** Control, Optimization, Reinforcement Learning, Machine Learning, Economics, Finance
- **Proficient in:** Python, PyTorch, R, Julia, Git, LATEX
- **Experience with:** Java, C, C++, MATLAB, Jira, Confluence
- **Relevant Courses:** Calculus I-IV, Linear Algebra, Probability, Statistics, Machine Learning, Time Series, Differential Equations, Control Theory, Numerical Optimization, OOP Software Engineering, Algorithms